

Xu Zhang

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EDUCATION

- **Illinois Institute of Technology** August 2024 - present
Doctor of Philosophy in Computer Engineering GPA: 4.00/4.00 Chicago, USA
- **Huazhong University of Science and Technology** August 2021 - June 2024
Master of Information and Communication Engineering GPA: 3.64/4.00 Wuhan, China
- **Huazhong University of Science and Technology** August 2017 - June 2021
Bachelor of Telecommunication Engineering GPA: 3.87/4.00 Wuhan, China

RESEARCH EXPERIENCE

- **Illinois Institute of Technology** August 2024 - present
Graduate Research Assistant, Trustworthy & Intelligent ML Group (Advisor: Prof. Ren Wang) Chicago, USA
 - **Multimodal LLM jailbreak attacks:** Proposed *Jailbreak Connectivity*, modeling adversarial images as a low-loss path for diverse, transferable, universal visual jailbreaks; **79.62%** avg. attack success rate on SafetyBench (**+36.24%** over the best baseline), transferring to GPT-4o and Gemini.
 - **In-context jailbreak analysis & defense:** Explained via posterior reweighting why harmful demonstrations break MLLM alignment (with scaling laws); built a risk-adaptive inference-time defense.
 - **Robust Mixture-of-Experts:** Showed experts are more vulnerable than the router; designed targeted robust training and a jointly trained dual-model with certified robustness bounds (ICML 2025, [code](#)).
 - **Multi-norm defense:** Proposed Robust CurveMoE, building experts along a mode-connectivity path; **+2.37/+2.13** pp Union accuracy over the best baseline on CIFAR-100/ImageNet-100.
- **Huazhong University of Science and Technology** September 2022 – June 2023
Teaching Assistant – Next-generation Wireless Network Technologies Wuhan, China
- **5G Verticals Innovation Laboratory** July 2019 - April 2021
Research Assistant Wuhan, China

PUBLICATIONS

Conference Papers

- [1] **Zhang X**, Xu K, Hu Z, Wang R. Optimizing Robustness and Accuracy in Mixture of Experts: A Dual-Model Approach. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, PMLR, 2025.  

Under Review

- [1] **Zhang X**, Hu Z, Han S, Wang R. Jailbreak Connectivity: Towards Diverse, Transferable, and Universal Multimodal Large Language Model Jailbreak. *Under review*, 2026.
- [2] **Zhang X**, Mistry D, Xu X, Wang R. Understanding In-Context Multimodal Jailbreaks via Posterior Reweighting. *Under review at ICLR 2027*. 
- [3] **Zhang X**, Wang R. Robust CurveMoE: Multi-Norm Adversarial Defense for Mixture-of-Experts Models via Mode Connectivity. *Under review at IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*. 

Workshop Papers

- [1] **Zhang X**, Hu Z, Han S, Wang R. Geometry-Driven Diverse and Transferable Visual Attacks on Multimodal LLMs. In *ICLR 2026 Workshop on Geometry-grounded Representation Learning and Generative Modeling*, 2026. 
- [2] Xiao Y, **Zhang X**, Li Y, Shi G, Başar T. Rate-Distortion Theory for Strategic Semantic Communication. In *Proceedings of the 2022 IEEE Information Theory Workshop (ITW)*, IEEE, 2022. 

SKILLS

- **Programming & Tools:** Python, PyTorch, Hugging Face Transformers, Git, Linux, LaTeX
- **Models:** LLaVA, MiniGPT-4, Qwen3-VL (dense & MoE), GPT-4o / Gemini APIs, ViT, ResNet / WideResNet
- **AI Safety & Robustness:** MLLM jailbreak & red teaming, adversarial attacks (PGD, AutoAttack), adversarial training, inference-time defenses, safety evaluation (MM-SafetyBench, AdvBench, Llama-Guard-3)

HONORS AND SERVICE

- **National Scholarship (Graduate, top 1%)**, Ministry of Education, China  October 2022
- **National Scholarship (Undergraduate, top 1%)**, Ministry of Education, China  October 2019
- **Reviewer**, ICML 2026; ICLR 2026 Workshop GRaM 2026